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$$y \in \{0, 1\}$$

$$\text{Linear score: } z = w^T x + b = \sum w_j x_j + b$$

$$\text{Hypothesis: } \hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$J(w) = \left(-\frac{1}{n}\right) \sum_{i=1}^n [y^{(i)} \log \hat{y}^{(i)} + (1 - y^{(i)}) \log (1 - \hat{y}^{(i)})]$$

$$l = - [y \log \hat{y} + (1 - y) \log (1 - \hat{y})]$$

$$\sigma(z) = (1 + e^{-z})^{-1}$$

$$\frac{\partial \sigma}{\partial z} = (1 + e^z)^{-2} \cdot (-e^{-z}) = \frac{e^{-z}}{(1 + e^{-z})^2}$$

$$\frac{e^{-z}}{1 + e^{-z}} = 1 - \sigma(z)$$

$$\frac{1}{1 + e^z} = \sigma(z)$$

$$\sigma'(z) = \sigma(z)(1 - \sigma(z)) = \hat{y}(1 - \hat{y})$$





$$\frac{dL}{du_j} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dz} \cdot \frac{dz}{du_j}$$

$$\frac{dz}{du_j} = X_j$$

$$\frac{d\hat{y}}{dz} = \hat{y}(1-\hat{y})$$

$$l = -y \log \hat{y} - (1-y) \log (1-\hat{y})$$

$$\frac{dL}{d\hat{y}} = \frac{-y}{\hat{y}} - (1-y) \cdot \frac{(-1)}{(1-\hat{y})} = \frac{-y}{\hat{y}} + \frac{(1-y)}{(1-\hat{y})}$$

$$\begin{aligned} \frac{dL}{d\hat{y}} &= \frac{[-y(1-\hat{y}) + (1-y)\hat{y}]}{[\hat{y}(1-\hat{y})]} = \frac{-y + y\hat{y} + \hat{y} - y\hat{y}}{\hat{y}(1-\hat{y})} \\ &= \frac{(\hat{y} - y)}{\hat{y}(1-\hat{y})} \end{aligned}$$

$$\frac{dL}{dy_j} = \left[(\hat{y} - y) / (\hat{y}(1-\hat{y})) \right] \cdot [\hat{y}(1-\hat{y})] \cdot x_j$$

$$= (\hat{y} - y) \cdot x_j$$

$$\frac{dJ}{dy_j} = \frac{1}{n} \sum_{i=1}^n (\hat{y}^{(i)} - y^{(i)}) x_j^{(i)}$$

